KERN 770-60
Data Output and Input Formats

Manual sections 6.4–6.5 – pages 116–121

6.4 Data Output Formats

Depending on the menu code setting:

- Code 7 2 1 = without data ID code
- Code 7 2 2 = with data ID code

Data will be output with either 16 characters (Code 7 2 1) or 22 characters (Code 7 2 2). For data output of 22 characters, a 6-character ID precedes the 16 characters reserved for the weight or other value.

Data Output Format with 16 Characters

Display segments that are not activated ("+" or "-" sign, leading zeros other than zeros before the decimal point) are output as spaces. The following data block format is output according to what is displayed on the balance:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Sign	Space	Weight	Weight	Weight	Weight	Weight	Weight	Weight	Weight	Space	U	U	U	CR	LF

* = space, U = unit

When data are output without decimals, the decimal point is suppressed (except when a certain display mode is selected).

Data output example: + 12.5557 g

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
+			1	2	.	5	5	5	7		g			CR	LF

Characters

Character(s)	Meaning
1st	Plus or minus sign or space
2nd	Space
3rd–10th	Weight with a decimal point; leading zeros = space
11th	Space
12th–14th	Unit symbol or space
15th	Carriage Return (CR)
16th	Line Feed (LF)

If the weighing system has not been stabilised, no unit symbol will be output.

Unit symbols

Symbol	Meaning	Symbol	Meaning
***	No stability parameter	GN*	Grains
o**	Grams (g)	dwt	Pennyweights
g**	Grams	mg*	Milligrams
kg*	Kilograms	/lb	Parts per Pound
ct*	Carats	tlc	Chinese taels
lb*	Pounds	mom	Mommes
oz*	Ounces	K**	Austrian carats
ozt	Troy Ounces	tol	Tola
tlh	Hong Kong taels	ba*	Baht
tls	Singapore taels	MS*	Mesghal
tlt	Taiwanese taels		

Special Codes

Special codes are output only if the balance operating menu code 6 1 1, 6 1 4 or 6 1 5 is set (see the section entitled “Data Output Parameters”).

Special status-dependent codes

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
*	*	*	*	*	*	A	B	*	*	*	*	*	*	CR	LF

The following status codes are output for “A B”:

- ** : Tare
- C* : Adjust[1]
- -- : All numerals indicated in stable readout
- H* : Overload
- L* : Underload

Special error-dependent codes

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
*	*	*	E	R	R	*	X	Y	Z	*	*	*	*	CR	LF

X = *, 0, 1, 2 as a one-place error code

YZ = two-place error index code

* = space

[1] For balances with a built-in automatic adjusting function, the displayed status code “C” will also be output when a print command is received.

6.5 Data Input Formats

Commands can be input via the balance interface port to control the balance functions.
Control commands are distinguished according to those with upper-case letters, or special characters, and those with lower-case letters.

Format for Control Commands

Control commands can include up to 13 characters.
Each character must be transmitted with a start bit, a 7-bit ASCII-coded character, a parity bit, and one or two stop bits.
You can define the parity, baud rate, handshake mode, and the number of stop bits by programming the respective codes in the balance operating menu.

Formats:

ESC	K	CR	LF

ESC	f	x	_	CR	LF

ESC = Escape (ASCII 27)
K, f = Command character (see next page)
X = Number
_ = Underline (ASCII 95)
CR = Carriage Return (ASCII 13)
LF = Line Feed (ASCII 10)

The characters CR and LF do not have to be transmitted in the data string.

Control Commands with Upper-case Letters or Special Characters

ESC	P	CR	LF	PRINT (print, activate/block auto print)
ESC	S	CR	LF	Restart/self-test
ESC	T	CR	LF	Tare
ESC	Z	CR	LF	Internal adjusting*

The P, T and Z commands do not affect the menu code settings of the balance. The S command causes the processor to reinitialise (turns the balance off and back on again).

The balances will operate according to the commands available up until the processor is reinitialised. Once the balance has been turned on, the processor will always recognise the codes entered by the user in the balance operating menu.

ESC	O	CR	LF	Block the keys
ESC	R	CR	LF	Release the keys

Important Note!
The PRINT key is not blocked!

Adaptation to Ambient Conditions

ESC	K	CR	LF	Very stable
ESC	L	CR	LF	Stable
ESC	M	CR	LF	Unstable
ESC	N	CR	LF	Very unstable

* = only for balances with a built-in automatic adjusting function

Source: KERN 770/GS/GJ manual – sections 6.4–6.5, pages 116–121